1 Vectors and Geometric Applications

Central question. How do we describe a quantity that has both a size and a direction?

1.1 From coordinates to directed displacements

In the first half of the course we used $(\cos\theta, \sin\theta)$ to locate a point on a circle. A vector uses the same coordinates to describe a *displacement*: how far to move horizontally and vertically. Write $\mathbf{v} = \langle a, b\rangle$. Its magnitude is

$$\boxed{\|\mathbf{v}\| = \sqrt{a^2 + b^2}}.$$

Vectors with the same components represent the same displacement even when drawn at different starting points.

Example 1.1. A walker moves 3 km east and 4 km north. The resulting vector is $\langle 3, 4\rangle$ km and its magnitude is 5 km. The route traveled is 7 km, while the displacement is only 5 km.

1.2 Vector arithmetic

Addition combines displacements:

$$\langle a, b\rangle + \langle c, d\rangle = \langle a + c, b + d\rangle.$$

Scalar multiplication changes their length, and a negative scalar reverses direction:

$$k\langle a, b\rangle = \langle ka, kb\rangle.$$

Subtracting vectors compares locations. If $P = (p_1, p_2)$ and $Q = (q_1, q_2)$, then $\overrightarrow{PQ} = \langle q_1 - p_1, q_2 - p_2\rangle$.

Example 1.2. For $\mathbf{u} = \langle 2, -1\rangle$ and $\mathbf{v} = \langle -3, 4\rangle$, we have $\mathbf{u} + \mathbf{v} = \langle -1, 3\rangle$ and $2\mathbf{u} - \mathbf{v} = \langle 7, -6\rangle$.

1.3 Direction and trigonometry

A unit vector has magnitude 1. The unit vector pointing at angle θ is

$$\mathbf{e}_\theta = \langle \cos\theta, \sin\theta\rangle.$$

A vector of magnitude r in that direction is

$$\boxed{\mathbf{v} = r\langle \cos\theta, \sin\theta\rangle}.$$

Conversely, for a nonzero vector $\langle a, b\rangle$, the angle must be chosen in the appropriate quadrant. An inverse tangent applied only to b/a may miss that quadrant.

Example 1.3. A force of 20 N directed 30° above the horizontal has components $\langle 10\sqrt{3}, 10\rangle$ N.

1.4 The dot product measures alignment

Define

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2.$$

The geometric interpretation is

$$\boxed{\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\|\|\mathbf{v}\| \cos\phi},$$

where ϕ is the angle between nonzero vectors. Positive dot product means an acute angle; negative means an obtuse angle; zero means perpendicular vectors.

Example 1.4. With $\mathbf{u} = \langle 3, 4\rangle$ and $\mathbf{v} = \langle 4, -3\rangle$, the dot product is $12 - 12 = 0$. Both vectors are nonzero, so they are perpendicular.

1.5 Projection and useful components

The scalar component of $\mathbf{u}$ in the direction of a nonzero vector $\mathbf{v}$ is

$$\operatorname{comp}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|}.$$

The corresponding vector projection is

$$\operatorname{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|^2}\mathbf{v}.$$

Projection is the part of one vector aligned with another. In physics, the work done by a constant force $\mathbf{F}$ across displacement $\mathbf{d}$ is $W = \mathbf{F} \cdot \mathbf{d}$, measured in joules when force is in newtons and displacement in meters.

Example 1.5. A force $\langle 10, 5\rangle$ N acts during displacement $\langle 3, 0\rangle$ m. The work is 30 J: only the horizontal force component contributes.

1.6 A geometric connection to the Law of Cosines

For $\mathbf{u}, \mathbf{v}$ we have

$$\|\mathbf{u} - \mathbf{v}\|^2 = (\mathbf{u} - \mathbf{v}) \cdot (\mathbf{u} - \mathbf{v}) = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\|\mathbf{u}\|\|\mathbf{v}\| \cos \phi.$$

Thus the Law of Cosines from Module 5 is also a theorem about vector differences.

1.7 Exercises

1. Find the vector from $P = (-2, 3)$ to $Q = (5, -1)$ and its magnitude.
2. For $\mathbf{u} = \langle 2, 5\rangle$, $\mathbf{v} = \langle -1, 3\rangle$, compute $\mathbf{u} + \mathbf{v}$ and $3\mathbf{u} - 2\mathbf{v}$.
3. Find exact components of a displacement of magnitude 12 at angle 150°.
4. Find a unit vector in the direction of $\langle -3, 4\rangle$.
5. Compute $\langle 2, -1\rangle \cdot \langle 5, 3\rangle$ and interpret its sign.
6. Determine whether $\langle 6, 2\rangle$ and $\langle -1, 3\rangle$ are perpendicular.
7. Find $\operatorname{proj}_{\mathbf{v}} \mathbf{u}$ when $\mathbf{u} = \langle 4, 3\rangle$ and $\mathbf{v} = \langle 1, 0\rangle$.
8. A force of 50 N acts 60° from the direction of a 4 m displacement. Calculate the work done.